

313 HWA

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Chapter 1

Project Statement

I chose the Leontief Input-Output model problem, which models the interdependence between 5 sectors of a national economy: Energy, Agriculture, Manufacturing, Transportation, and Services. Each sector both consumes and supplies outputs to the others. The goal is to determine the total level of production required to meet both internal and consumer demand.

Chapter 2

Mathematical Formulation

2.1 Variables

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} NRG \\ ARG \\ MAN \\ TRN \\ SRV \end{bmatrix}$$

2.2 Economic Principle

Each sector in the Economy both consumes and produces goods. A portion of each sector's output is used internally by other sectors as inputs (intermediate demand), while the rest satisfies final demand (consumers, gov., or export demand). Therefore, the total production of each sector must equal the sum of:

1. "Internal demand (goods used by other sectors) - represented by Cx "
2. "Final demand (goods demanded by consumers) - represented by d "

This gives the Leontief Input-Output equation:

$$x = Cx + d$$

2.3 Standard Linear Algebra Form

We want to rearrange this equation into our usual $Ax = b$ equation. So first subtract Cx from both sides:

$$x - Cx = d$$

Then we factor out x :

$$(I - C)x = d$$

Where I is the 5×5 identity matrix:

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

So the linear system is:

$$A = (I - C), b = d$$

And we solve

$$Ax = b$$

2.4 Explicit System

We can write out $I - C$ numerically:

$$I - C = \begin{bmatrix} 1 - 0.30 & 0 - 0.15 & 0 - 0.20 & 0 - 0.30 & 0 - 0.10 \\ 0 - 0.05 & 1 - 0.20 & 0 - 0.25 & 0 - 0.00 & 0 - 0.05 \\ 0 - 0.15 & 0 - 0.10 & 1 - 0.25 & 0 - 0.20 & 0 - 0.15 \\ 0 - 0.05 & 0 - 0.10 & 0 - 0.10 & 1 - 0.15 & 0 - 0.10 \\ 0 - 0.10 & 0 - 0.05 & 0 - 0.10 & 0 - 0.10 & 1 - 0.20 \end{bmatrix}$$

$$I - C = \begin{bmatrix} 0.70 & -0.15 & -0.20 & -0.30 & -0.10 \\ -0.05 & 0.80 & -0.25 & 0.00 & -0.05 \\ -0.15 & -0.10 & 0.75 & -0.20 & -0.15 \\ -0.05 & -0.10 & -0.10 & 0.85 & -0.10 \\ -0.10 & -0.05 & -0.10 & -0.10 & 0.80 \end{bmatrix}$$

and

$$d = \begin{bmatrix} 80 \\ 60 \\ 150 \\ 75 \\ 200 \end{bmatrix}$$

Chapter 3

Software Implementation

```

Command Window
>> C = [0.3,0.15,0.2,0.3,0.1;0.05,0.2,0.25,0,0.05;0.15,0.1,0.25,0.2,0.15;0.05,0.1,0.1,0.15,0.1;0.1,0.05,0.1,0.1,0.2]

C =

    0.3000    0.1500    0.2000    0.3000    0.1000
    0.0500    0.2000    0.2500         0    0.0500
    0.1500    0.1000    0.2500    0.2000    0.1500
    0.0500    0.1000    0.1000    0.1500    0.1000
    0.1000    0.0500    0.1000    0.1000    0.2000

>> d = [80;60;150;70;200]

d =

    80
    60
   150
    70
   200

>> I = eye(5)

I =

     1     0     0     0     0
     0     1     0     0     0
     0     0     1     0     0
     0     0     0     1     0
     0     0     0     0     1

>> A = I-C

A =

    0.7000   -0.1500   -0.2000   -0.3000   -0.1000
   -0.0500    0.8000   -0.2500         0   -0.0500
   -0.1500   -0.1000    0.7500   -0.2000   -0.1500
   -0.0500   -0.1000   -0.1000    0.8500   -0.1000
   -0.1000   -0.0500   -0.1000   -0.1000    0.8000

```

Figure 3.1: MATLAB implementation of the Leontief Input-Output model 1.

```

A =

    0.7000    -0.1500    -0.2000    -0.3000    -0.1000
   -0.0500     0.8000    -0.2500         0    -0.0500
   -0.1500    -0.1000     0.7500    -0.2000    -0.1500
   -0.0500    -0.1000    -0.1000     0.8500    -0.1000
   -0.1000    -0.0500    -0.1000    -0.1000     0.8000

>> x = A\d; %solve for x
>> Cx = C*x %internal demand

Cx =

    400.4120
    222.6205
    334.3952
    180.2073
    219.5406

>> x

x =

    480.4120
    282.6205
    484.3952
    250.2073
    419.5406

```

Figure 3.2: MATLAB implementation of the Leontief Input-Output model 2.

Chapter 4

Results and Interpretation

4.1 Results

Total production (In billions of dollars worth of goods):

$$x = \begin{bmatrix} 480.4120 \\ 282.6205 \\ 484.3952 \\ 250.2073 \\ 419.5406 \end{bmatrix}$$

Internal demand (In billions of dollars worth of goods):

$$Cx = \begin{bmatrix} 400.4120 \\ 222.6205 \\ 334.3952 \\ 180.2073 \\ 219.5406 \end{bmatrix}$$

4.2 Relationship Verification

$$x_{MAN} = (Cx)_{MAN} + d_{MAN}$$

$$x_{MAN} = 484.3952$$

$$Cx_{MAN} = 334.3952, d_{MAN} = 150$$

$$334.3952 + 150 = 484.3925 = x_{MAN}$$

4.3 Interpretation

- x_i is the total required production in each sector (in billions of dollars).
- $(Cx)_i$ is the internal demand (how is consumed by other sectors).
- d_i is the final demand (external/consumer demand)

4.4 Summary

The model predicts that the total production required to satisfy all internal and external demand is:

$$x = \begin{bmatrix} 480.4120 \\ 282.6205 \\ 484.3952 \\ 250.2073 \\ 419.5406 \end{bmatrix}$$

This means that, for example, the manufacturing sector must produce \$484.3952 billion worth of goods, of which \$334.3952 billion will be consumed internally (by other industries), and \$150 billion will go to final consumers.